Atharva Kirkole

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EDUCATION

Accelerated path: completed B.S. in 3 years + M.S. in 1 year — both in Computer Science.

Michigan State University — East Lansing, MI
M.S. Computer Science — GPA 3.93 | Teaching Assistant, OOP, Parallel Programming — *Aug 2025 – May 2026*

Michigan State University — East Lansing, MI
B.S. Computer Science, Minor in Business (Honors College) — GPA 3.93, Honors, Tau Beta Pi — *Aug 2022 – Apr 2025*

EXPERIENCE

Heard AI — East Lansing, MI
Speech AI Research Assistant — *Mar 2026 – June 2026*

- Contributed to $5M NSF Convergence Accelerator (Phase 2) project on accessible speech technologies; developed speaker diarization and stutter detection models to identify and classify dysfluent speech patterns.
- Built initial stutter speech synthesis/regeneration pipeline to generate fluent speech alternatives from stuttered input; engineered full-stack application (Node.js, React, AWS Cloudscape) serving 300 concurrent sessions at <400ms latency.
- Implemented per-user fine-tuning framework achieving 23% WER improvement (14% → 10.8%) using rapid adaptation.

SPaRTa Lab (Scalable Parallel Technologies), MSU — East Lansing, MI
High Performance Computing Research Assistant — *Jun 2026 – Present*

- Parallelized the linearized Stokes equation over 10M+ sphere- and ellipsoid-surface points, simulating rigid bodies in viscous (low-Reynolds) flow — with applications in red-blood-cell dynamics and drug-discovery modeling.
- Built a CUDA boundary-integral solver reaching 60× speedup via coalesced memory access and shared-memory tiling; applied numerical analysis and error-control techniques to n-body interactions, reducing numerical error from 7% to 2%.

Michigan State University IT Services — East Lansing, MI
Software Developer — *Jun 2024 – Apr 2026*

- Led end-to-end migration of the campus mapping platform off Google Maps onto a self-managed ESRI ArcGIS stack serving **100K+ daily requests**, rebuilding map surface around custom geospatial layers and cutting licensing cost ~20% (**~$45K/yr**).
- Owned geospatial data modeling for ~12K+ campus entities/POIs; built an automated telemetry ingestion pipeline processing real-time location signals to drive data-driven route optimization, cutting compute latency ~30% (~210ms → ~145ms).
- Engineered student/people information-system APIs and Dockerized service images (PHP, JavaScript); raised site accessibility from 80% to **99.7%** (WCAG 2.1 / ADA / ARIA) across **40** production surfaces.

Amazon — East Lansing, MI
University Capstone Project — *Jan 2025 – May 2025*

- Built an AI-powered semantic code-search system over **~10k** internal architecture assets using AWS OpenSearch (Elasticsearch-compatible), Lambda, DynamoDB, and S3.
- Designed a RAG-style retrieval pipeline (dynamically chunked vector embeddings via Bedrock + SageMaker, indexed in OpenSearch) at **~90ms**; tuned indexing/ranking to lift top-k relevance **~18%** (NDCG@10) and added vector-similarity employee recommendations over GitHub signals.

Thompson Maize Lab — East Lansing, MI
Machine Learning Research Intern — *Aug 2023 – May 2024*

- Built and evaluated Random Forest / ML models (TensorFlow, Python, R) for crop-yield prediction, querying 120+ dimensional agronomic datasets from PostgreSQL with relational joins/aggregations (~72% accuracy); presented at NAPPN Conference, Purdue, with a published abstract (link).

PROJECTS

FitTrack — Wearable Time-Series Analytics & Health Prediction (Python / TensorFlow) — 2025

- Built a real-time fitness analytics platform processing high-volume wearable sensor streams (~50Hz);
- implemented streaming data pipelines, time-series SQL storage, and a live metrics dashboard.
- Trained ML models predicting **age, gender, BMI, and body-fat %** from sensor signals — **~88%** accuracy, **~3.5% MAE** on body-fat regression across **~1.5M** sample windows.

NanoGPT Attention Kernel Optimization (CUDA / C++17) — Jan 2026

- Implemented Transformer multi-head attention in C++17 over B, H, N, D tensors with a 4D accessor collapsing index/stride arithmetic; row-tiled $Q \cdot K^T$ and $P \cdot V$ matmuls to keep K/V panels resident in L1/L2.
- Parallelized across (batch, head) with OpenMP, vectorized with AVX2 + FMA, and fused matmul–softmax–matmul for a 68× speedup; built a CUDA variant with cooperative shared-memory tiling and per-thread online-softmax state.

Distributed Systems(MPI) – Masters Parallel Computing Course (4.0) — Jan 2026

- Latency hiding: overlapped non-blocking communication (Isend/Irecv) with stencil computation
- One-sided RMA: MPI_Win/MPI_Get for direct array access; performance gains on large datasets (100M+ elements/rank)
- Custom data types: MPI_Type_create_struct with offsetof for efficient MPI_Gather & Subcommunicators: MPI_Comm_split

TECHNICAL SKILLS

Maps & Geospatial: ESRI ArcGIS, ArcGIS Portal, Google Maps API, Leaflet.js, GIS data modeling, POI/entity modeling
Tools: Hadoop, AWS, OpenSearch, Bedrock, SageMaker, EC2, RAG, time-series analytics, DynamoDB, MongoDB, S3
Languages & ML/Web: C++17, C#, Python, Telemetry, JavaScript, SQL, R, PHP; PyTorch, TensorFlow, React, Node.js, Flask, Django, .NET, REST APIs, Docker, AWS Lambda; CUDA / OpenMP / SIMD, PostgreSQL

CERTIFICATIONS

Graduate Certificate in High-Performance Computing, MSU • CS50x, HarvardX (Verified).